

Deterministic Learning-Based Fault Identification for Nonlinear Sampled-Data Systems: Learning Accuracy Analysis

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Abstract—In this article, a sampled-data fault identification (SDFI) scheme for nonlinear uncertain systems is proposed based on the deterministic learning approach, and the learning performance of the SDFI algorithm is analyzed. First, a learning-based estimator is designed, and a sampled-data (SD) linear time-varying (LTV) system is employed to describe the learning systems. Second, by constructing a special time-varying symmetric positive definite matrix, the exponential convergence property of the SD LTV system is derived. Third, explicit formulas for learning accuracy are established, which illuminate the relation between the learning performance, persistent excitation (PE) level of neural networks (NNs), and the parameters of the learning system. The main advantage of the theoretical results given in this article is that the parameters in learning accuracy formulas can be derived from measurable signals, allowing for effective evaluation of the learning performance of the SDFI scheme in practical applications. Simulation studies of a robot manipulator and a compressor system are included to show the effectiveness of the proposed method.

Index Terms—Deterministic learning, fault identification (FI), persistent excitation (PE) condition, sampled-data (SD) systems.

I. INTRODUCTION

WITH the rapid development of modern industry, there is an increasing demand for safety and reliability in industrial systems [1]. Failure to detect faults in a timely manner will cause inevitable losses or even catastrophes. Therefore, fault diagnosis (FD) of industrial systems has become a crucial task.

In the past decades, a large number of FD methods using various techniques have been developed, which can be divided into two main categories: model-based and data-based methods [2], [3]. The first category of FD methods requires obtaining a model of a dynamical process. However, in many practical applications, there exist a lot of uncertainties that are difficult to obtain through physical principles. The second category of FD methods involves implementing intelligence technologies

to extract useful fault features from massive amounts of data and then diagnose different faults, such as deep learning [4], transfer learning [5], reinforcement learning [6], [7], and adaptive learning [8], [9], [10].

Recently, considerable attention has been paid to developing FD and fault-tolerant control (FTC) methods with guaranteed performance. In [11], performance-based FD and FTC methods were proposed by applying the Takagi–Sugeno fuzzy dynamic modeling technique. In [12] and [13], the issues of fixed-time and predefined-time fault estimation were studied for complex networks and interconnected systems. In [14], a predefined-time observer is proposed for sector-bounded nonlinear systems to detect both the emergence and dissipation of intermittent faults. In order to enhance the accuracy and reliability of the fault signal estimation, an estimator based on an unknown-input observer was constructed to decouple the estimation error dynamics from the influence of fault signals [15]. FTC methods based on prescribed performance control approaches have been developed, in which the prescribed transient and steady-state performance can be guaranteed [16], [17], [18]. To achieve fast transient performance, finite-time control schemes have also been employed to address the FTC problem with finite-time prescribed performance [19], [20]. Most of the existing performance-based FD and FTC approaches focus on the diagnosis and control performance. Nevertheless, analysis of the learning performance of the neural networks (NNs) within FD/FTC schemes has received relatively less attention.

The objective of this article is to develop an NN-based fault identification (FI) scheme with guaranteed learning accuracy. These issues are important but difficult. The first difficulty is that it requires satisfying the persistent excitation (PE) condition so that the NN weights can be guaranteed to converge into their ideal values. Nevertheless, the PE condition is very restrictive. To overcome this difficulty, in [21] and [22], a theory named deterministic learning has been developed, in which it is proven that by using localized radial basis function (RBF) NNs, the periodic/quasi-periodic trajectories can lead to the satisfaction of the partial PE condition of the regressor vector consisting of the neurons along the trajectory. With the partial PE condition, the NN weights will converge into small neighborhoods of their ideal values, and the local accurate approximation of unknown system dynamics and fault dynamics can be achieved [21], [23], [24], [25]. The second

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difficulty is how to determine the convergence accuracy of NN weights. In [26] and [27], relations between disturbance, the PE condition (or PE level), and the sizes of convergence neighborhoods of NN weights have been investigated. Moreover, the relationship between PE levels and RBF network structures has been investigated [28]. However, due to the coupling between estimation and adaptation, performance analysis results given in [26] and [27] are complex and not easily applicable.

In this article, we develop a deterministic learning-based FI problem for the nonlinear sampled-data (SD) systems, and investigate the problem of how to derive an applicable quantitative analysis of the learning accuracy (the sizes of convergence neighborhoods of NN weights) for the proposed learning-based FI method. First, a learning-based estimator is designed, and an SD linear time-varying (LTV) system is employed to describe the learning system. Second, the exponential convergence property of the SD LTV system (learning system) is analyzed. Due to the interaction between system states and NN weights, a direct analysis of the convergence properties of the learning system is very complex. To overcome this difficulty, a time-varying positive definite matrix (TVPDM) is constructed, which is based on the state transition matrix of a subsystem that only consists of NN weight estimation. By exploiting the observability and exponential convergence property of this subsystem, the boundedness and convergence properties of the TVPDM are established. Then, the TVPDM is used to design a time-varying Lyapunov function to analyze the exponential convergence property of the learning systems. Third, learning accuracy formulas are derived, which characterize the sizes of convergence neighborhoods of NN weights.

Compared with the analysis results given in [26] and [27], the main advantage of the theoretical results given in this article is that the parameters in our learning accuracy formulas can be derived from measurable signals, allowing for effective evaluation of the learning accuracy of the FI scheme in practical applications. These results not only illuminate the relation between learning accuracy, PE level of NNs, and the parameters of the learning system but also verify whether the learning-based FI scheme can satisfy a given level of accuracy.

The rest of this article is organized as follows: the problem formulation and partial PE condition are introduced in Section II. Section III presents the process of FI and analyzes the learning accuracy. In Section IV, simulation studies of a robot manipulator and a compressor system are included to validate our scheme. Section V concludes this article.

II. PROBLEM FORMULATION AND PRELIMINARIES

A. Problem Formulation

A general nonlinear system is considered as follows:

$$\dot{x} = f(x, u) + \gamma(x, u) + \phi(x, u) + \omega(t) \quad (1)$$

where $x = [x_1, \dots, x_n]^T$ is the state vector of the system, $u = [u_1, \dots, u_m]^T$ represents the control input vector, and $f: \mathbb{R}^n \times \mathbb{R}^m \mapsto \mathbb{R}^n$ and $\gamma: \mathbb{R}^n \times \mathbb{R}^m \mapsto \mathbb{R}^n$ are smooth vector fields. f represents the known nominal system dynamics. γ represents the unmodeled dynamics of the system and

$\omega(t) = [\omega_1, \dots, \omega_n]^T$ represents the noise and disturbances. $\phi: \mathbb{R}^n \times \mathbb{R}^m \mapsto \mathbb{R}^n$ denotes the deviation in the system dynamics caused by the unknown fault.

Selecting a sampling interval τ , we obtain that $x_k = x(k\tau)$, $u_k = u(t) \forall t \in [k\tau, (k+1)\tau)$. The Euler approximation for the SD representation of the nonlinear system described in (1) can be given as

$$x_{k+1} = x_k + \tau(f(x_k, u_k) + \gamma(x_k, u_k) + \theta_{k-K_0}\phi(x_k, u_k) + \omega_k) \quad (2)$$

where $\omega_k = \omega(t)$, K_0 denote the moment of fault occurrence.

Remark 1: In the Euler approximation, the selection of the sampling interval τ is crucial to the performance of identification schemes. A small sampling interval τ will improve the approximation accuracy but increase the computational load of learning schemes, whereas a large τ will reduce the computational load but degrade the accuracy of identification schemes. To take a tradeoff between these two issues, the sampling interval τ can be chosen according to the sampling theorem. That is, based on the frequencies of the fault dynamics, τ should be properly selected such that the signal is not distorted.

Assumption 1: Under fault modes, the system states and inputs remain bounded, that is, $(x_k, u_k) \in \Omega$, where $\Omega \subset \mathbb{R}^{n+m}$ is a compact set. Furthermore, the trajectory of the system is a periodic, periodic-like, or chaotic trajectory.

Remark 2: Periodic/periodic-like faults widely exist in practical engineering systems, particularly in domains like power grid systems [29], aero engines [30], and the chemical industry [31]. For example, in wind power grid-connected systems, oscillations in the grid may cause large-scale wind turbines to go off-grid [29]. Furthermore, oscillation within an aero-engine can give rise to various phenomena, such as rotating stall and surge, ultimately resulting in blade fracture and engine failure [32]. Therefore, the detection and identification of periodic/periodic-like faults are paramount in engineering systems.

Note that the modeling uncertainty $\gamma(x_k, u_k)$ and faults $\phi(x_k, u_k)$ are difficult to decouple from each other. The two terms will be treated as an inseparable term, denoted by $\xi(x_k, u_k) = \gamma(x_k, u_k) + \phi(x_k, u_k)$. Therefore, the system (2) in fault mode can be rewritten as follows:

$$x_{k+1} = x_k + \tau(f(x_k, u_k) + \xi(x_k, u_k) + \omega_k). \quad (3)$$

The objectives of this article are to identify the unmodeled fault dynamics and give a quantitative analysis of the learning accuracy for the proposed FI method.

B. Localized RBF Networks and PE

The RBF networks can be characterized by

$$F_m(V) = W^T S(V) = \sum_{i=1}^Q w_i s_i(V) \quad (4)$$

where $V \in \Omega_V \subset \mathbb{R}^q$ is the input vector, $W = [w_1, \dots, w_Q]$ is the weight vector, and Q is the number of NN nodes. Then, the RBF can be expressed as $S(V) = [s_1(V), \dots, s_Q(V)]^T$, in which the Gaussian basis function is usually chosen.

As shown in [21], for a trajectory $\psi_\zeta \in \Omega_V$ and a continuous function $F(V) : \Omega_V \rightarrow \mathbb{R}$, $F(V)$ can be approximated by the neurons along the trajectory ψ_ζ , i.e.,

$$F(V) = W_\zeta^{*T} S_\zeta(V) + \epsilon(V) \quad (5)$$

where $S_\zeta(Z) = [s_{j_1}(V), \dots, s_{j_\zeta}(V)]^T$ is the localized RBF vector that consists of the neurons along the trajectory ψ_ζ , $W^* = [w_{j_1}^*, \dots, w_{j_\zeta}^*]^T$ is the corresponding ideal weight vector, and $|\epsilon(V)| < \epsilon^*$ with ϵ^* is tolerable error.

Moreover, the following lemma shows the partial PE property of RBF networks.

Lemma 1 ([21], [33]): Assuming the input vector V_k of the RBF network is recurrent and within a compact set Ω_V , the regressor subvector $S_\zeta(V_k)$ will satisfy the PE condition. That is, there exists positive constants α_1 , α_2 , and δ such that

$$\alpha_2 I > \sum_{k=k_0}^{k_0+\delta} S_\zeta(V_k) S_\zeta(V_k)^T > \alpha_1 I \quad \forall k_0 > 0. \quad (6)$$

Definition 1 (PE Levels): Zheng and Wang [28] α_1 , α_2 in (6) refer to the level of excitation and the upper bound of excitation, respectively, and are collectively called the PE levels.

III. FI AND CONVERGENCE ACCURACY ANALYSIS

In this section, we present a learning estimator that uses deterministic learning to identify unknown faults in the SD system. Then, we give the convergence accuracy analysis for the FI scheme.

The following learning estimator is exploited to acquire the fault dynamics of system (3):

$$\begin{aligned} \hat{x}_{i(k+1)} &= x_{ik} + a_i (\hat{x}_{ik} - x_{ik}) \\ &\quad + \tau (f_i(x_k, u_k) + \hat{W}_{i\zeta k}^T S_{i\zeta}(x_k, u_k)) \end{aligned} \quad (7)$$

where x_{ik} is the state from the system (3), $\hat{x}_{ik}$ is the estimate of x_{ik} , a_i is the estimator gain, f_i is the known nominal system dynamics, and $\hat{W}_{i\zeta k}^T S_{i\zeta}(x_k, u_k)$ is utilized to approximate the unknown function $\xi_i(x_k, u_k)$, $i = 1, \dots, n$. Using the local approximation property of RBF NN (5), $\hat{W}_{i\zeta k}^T S_{i\zeta}(x_k, u_k)$ is constructed by properly selecting the neurons along the system trajectory.

Define

$$\begin{aligned} e_{ik} &= \hat{x}_{ik} - x_{ik} \\ \tilde{W}_{i\zeta k} &= \hat{W}_{i\zeta k} - W_{i\zeta}^* \end{aligned} \quad (8)$$

where $W_{i\zeta}^*$ is the ideal weight vector, $\hat{W}_{i\zeta k}$ is the estimate of $W_{i\zeta}^*$, and e_{ik} and $\tilde{W}_{i\zeta k}$ are the tracking errors of the states and weights, respectively.

According to (7) and (8), we have

$$\begin{aligned} e_{i(k+1)} &= a_i e_{ik} + \tau (\hat{W}_{i\zeta k}^T S_{i\zeta k} - \xi_{ik} - \omega_{ik}) \\ &= a_i e_{ik} + \tau (\tilde{W}_{i\zeta k}^T S_{i\zeta k} - \bar{\omega}_{ik}) \end{aligned} \quad (9)$$

where $S_{i\zeta k} = S_{i\zeta}(x_k, u_k)$ and $\xi_{ik} = \xi_i(x_k, u_k)$, $\tilde{W}_{i\zeta k} = \hat{W}_{i\zeta k} - W_{i\zeta}^*$, $\bar{\omega}_{ik} = \epsilon_{ik} + \omega_{ik}$, ϵ_{ik} is tolerable error defined in (5) and $\omega_i(k)$ is the noise and disturbances. The adaptive law for the NN weight is designed as

$$\hat{W}_{i\zeta(k+1)} = \hat{W}_{i\zeta k} - \Gamma_i S_{i\zeta k} e_{i(k+1)} \quad (10)$$

where $\Gamma_i > 0$ is a constant adaptation gain.

Remark 3: The adaptive law for NN weights is designed based on the state-prediction error $e_{i(k+1)}$. Note that $e_{i(k+1)} = x_{i(k+1)} - \hat{x}_{i(k+1)}$. $x_{i(k+1)}$ is the true system state, which is available from A/D sensor converters at the sampling instant $(k+1)\tau$. $\hat{x}_{i(k+1)}$ is the estimator state, which can be computed by (7), using the previous state x_{ik} , $\hat{x}_{ik}$, input u_{ik} , and NN weight $\hat{W}_{i\zeta k}$. Therefore, $e_{i(k+1)}$ is available at the sampling instant $(k+1)\tau$, and then can be used to update the NN weight $W_{i\zeta(k+1)}$.

Then, subtracting $W_{i\zeta}^*$ from both sides of (10), and substituting e_{ik} from (9) into (10), it can be converted to

$$\begin{aligned} \tilde{W}_{i\zeta(k+1)} &= \tilde{W}_{i\zeta k} - \Gamma_i a_i S_{i\zeta k} e_{ik} - \Gamma_i \tau S_{i\zeta k} \tilde{W}_{i\zeta k}^T S_{i\zeta k} \\ &\quad + \Gamma_i \tau S_{i\zeta k} \bar{\omega}_{ik}. \end{aligned} \quad (11)$$

From (9) and (11), the learning error system can be written in the LTV form as follows:

$$\begin{aligned} \begin{bmatrix} e_{i(k+1)} \\ \tilde{W}_{i\zeta(k+1)} \end{bmatrix} &= \begin{bmatrix} a_i & \tau S_{i\zeta k}^T \\ -a_i \Gamma_i S_{i\zeta k} & I - \tau \Gamma_i S_{i\zeta k} S_{i\zeta k}^T \end{bmatrix} \begin{bmatrix} e_{ik} \\ \tilde{W}_{i\zeta k} \end{bmatrix} \\ &\quad + \begin{bmatrix} \tau \bar{\omega}_{ik} \\ \Gamma_i \tau S_{i\zeta k} \bar{\omega}_{ik} \end{bmatrix}. \end{aligned} \quad (12)$$

In the following, the learning accuracy of the proposed method is analyzed. To address this issue, the following lemmas are needed. The first lemma shows uniformly observability of the SD LTV system with output injection, which is extended from [34, Lemma 2.5.2]

Lemma 2: Consider the LTV system defined by

$$\begin{aligned} x_{k+1} &= A_k x_k \\ y_k &= C_k x_k \end{aligned} \quad (13)$$

and the LTV system under output injection

$$\begin{aligned} w_{k+1} &= (A_k + K_k C_k) w_k \\ y'_k &= C_k w_k. \end{aligned} \quad (14)$$

Assume that, for all $\delta \geq 0$, there exists $k_\delta \geq 0$ such that, for all $k_0 \geq 0$

$$\sum_{i=k_0}^{k_0+\delta} |K_i|^2 \leq k_\delta. \quad (15)$$

Then, the system $[C_k, A_k]$ is uniformly observable if and only if the system $[C_k, A_k + K_k C_k]$ is uniformly observable. Moreover, if there exist constants $\beta_1 > 0$, $\beta_2 > 0$, and some integer $\delta > 0$ such that the output of the system $[C_k, A_k]$ satisfies

$$\beta_2 |x_{k_0}|^2 \geq \sum_{k=k_0}^{k_0+\delta} y_k^T y_k \geq \beta_1 |x_{k_0}|^2. \quad (16)$$

Then, the output of the system $[C_k, A_k + K_k C_k]$ satisfies

$$\sum_{k=k_0}^{k_0+\delta} y_k'^T y_k' \geq \beta'_1 |w_{k_0}|^2 \quad (17)$$

with

$$\beta'_1 = \frac{\beta_1}{(1 + \sqrt{k_\delta \beta_2})^2}. \quad (18)$$

Proof: Please see the Appendix for the details of the proof of Lemma 2. ■

Note that the LTV system (12) consists of the state and parameter vectors. Due to the coupling between estimation and adaptation, direct analysis of convergence properties of the LTV system (12) is very complex. In [35], a Lyapunov method with a TVPDM is designed for exponential convergence analysis for an adaptive observer algorithm in continue time case. Inspired by this technique, the Lyapunov method with a TVPDM is exploited to overcome the difficulty in convergence analysis of the LTV system (12).

Similar to [35], we consider the following system:

$$\begin{aligned} w_{k+1} &= (I - \tau \Gamma_i S_{i\zeta k} S_{i\zeta k}^T) w_k \\ y_k &= S_{i\zeta k}^T w_k. \end{aligned} \quad (19)$$

It can be proved that the above system is exponentially stable when $S_{i\zeta k}$ is PE [27]. Then, a classical stability result on LTV systems is exploited to derive the TVPDM, which is given as follows.

Lemma 3 [36]: Consider the following linear state equation:

$$x_{k+1} = A_k x_k.$$

The above system is exponentially stable if and only if there exists a symmetric matrix sequence $P(k)$ such that

$$\begin{aligned} \eta I &\leq P(k) \leq \rho I \\ A_k P(k+1) A_k^T - P(k) &\leq -\nu I \end{aligned}$$

where η , ρ , and ν are finite positive constants.

Therefore, based on the above lemma, there exists a symmetric positive definite matrix $P(k)$ such that

$$G_k^T P(k+1) G_k - P(k) \leq -\nu I \quad (20)$$

where $G_k = (I - \tau \Gamma_i S_{i\zeta k} S_{i\zeta k}^T)$.

Note that how to design the TVPDM is not given in [35]. To address this issue, inspired by the results in the proof of [34, Theorem 1.5.6], we define a TVPDM as follows:

$$P(k) = \sum_{\sigma=k}^{k+\delta-1} \Phi^T(\sigma, k) \Gamma_i^{-1} \Phi(\sigma, k) \quad (21)$$

where δ is a parameter given in (6) and $\Phi(\sigma, k)$ is the state transition matrix of the system (19), which is expressed by

$$\Phi(\sigma, k) = \begin{cases} \prod_{i=k}^{\sigma-1} G_i, & \sigma \geq k+1 \\ I, & \sigma = k. \end{cases} \quad (22)$$

Through exploiting the observability and exponential convergence property of the system (19), the following lemma provides the lower and upper bounds of $P(k)$ and the parameter ν in (3).

Lemma 4: Consider the matrix $P(k)$ defined by (21). If $\tau S_{i\zeta k}^T \Gamma_i S_{i\zeta k} < 2$ and $S_{i\zeta k}$ satisfies the condition (6), then we have

$$\text{i) } G_k^T P(k+1) G_k - P(k) \leq -\kappa \tau \lambda_1 I \quad (23)$$

$$\text{ii) } \frac{\lambda_2}{\Gamma_i} I \leq P(k) \leq \frac{\delta}{\Gamma_i} I \quad (24)$$

where

$$\kappa = 2 - \tau \Gamma_i \mu_3^2 \quad (25)$$

$$\mu_3 = \max_{k \geq k_0} \|S_{i\zeta k}\| \quad (26)$$

$$\lambda_1 = \alpha_1 / \left(1 + \sqrt{Q_\zeta} \Gamma_i \tau \alpha_2\right)^2 \quad (27)$$

$$\lambda_2 = \delta / (1 + \tau \Gamma_i \lambda_3 \delta)^2 \quad (28)$$

where Q_ζ is the dimension of vector $S_{i\zeta k}$, λ_3 is the maximum eigenvalue of $S_{i\zeta k} S_{i\zeta k}^T$ for all $k \geq k_0$, and k_0 is the initial time.

Proof: Please see the Appendix for the details of the proof of Lemma 4. ■

Let $\bar{\epsilon}_{nn}$ be the learning accuracy of the proposed SD fault identification (SDFI) scheme, which is defined as follows:

$$\bar{\epsilon}_{nn} := |\tilde{W}_{i\zeta k}^T S_{i\zeta k}|. \quad (29)$$

By using the TVPDM $P(k)$ given in (21) to design a Lyapunov function, the learning accuracy of the proposed SDFI scheme is derived as follows.

Theorem 1: Consider the learning error system (12) which satisfies Assumption 1. If the following conditions hold:

$$0 < \Gamma_i < (\tau \mu_3^2)^{-1} \quad (30)$$

$$a_i < \left(\frac{\tau}{\tau \mu_1 + \mu_2} \right)^{\frac{1}{2}}. \quad (31)$$

Then, the neural-weight estimate error $\tilde{W}_{i\zeta k}$ will converge into the following compact set:

$$\Omega_{\tilde{W}_{i\zeta k}} \triangleq \left\{ \tilde{W}_{i\zeta k} \mid \|\tilde{W}_{i\zeta k}\| \leq \sqrt{\frac{2(\mu_1 + \mu_2)}{\kappa \tau \lambda_1}} \omega_i^* \right\} \quad (32)$$

and the NN approximation accuracy satisfies

$$\bar{\epsilon}_{nn} \leq \sqrt{\frac{2(\mu_1 + \mu_2)}{\kappa \tau \lambda_1}} \mu_3 \omega_i^* \quad (33)$$

where

$$\mu_1 = (1 + \tau) / (1 - \tau \Gamma_i \mu_3^2) \quad (34)$$

$$\mu_2 = 2\delta \tau^2 \Gamma_i \mu_3^2 + 4\delta^2 \mu_3^2 \tau / (\kappa \lambda_1) \quad (35)$$

$$\omega_i^* = \sup_{k \geq k_0} \{|\bar{\omega}_{ik}|\} \quad (36)$$

where μ_3 , κ , and λ_1 are parameters given in Lemma 4, δ is a parameter given in (6), and $\bar{\omega}_{ik}$ is defined in (9).

Proof: Consider the following Lyapunov candidate function:

$$V_k = V_{ka} + V_{kb} \quad (37)$$

where $V_{ka} = \tau^{-1} e_{ik}^2 + \tilde{W}_{i\zeta k}^T \Gamma_i^{-1} \tilde{W}_{i\zeta k}$, $V_{kb} = \tilde{W}_{i\zeta k}^T P(k) \tilde{W}_{i\zeta k}$, $P(k)$ is defined in (21).

The forward difference of V_{ka} is derived as follows:

$$\begin{aligned} \Delta V_{ka} &:= V_{(k+1)a} - V_{ka} \\ &= \frac{1}{\tau} e_{i(k+1)}^2 + \tilde{W}_{i\zeta(k+1)}^T \Gamma_i^{-1} \tilde{W}_{i\zeta(k+1)} \\ &\quad - \frac{1}{\tau} e_{ik}^2 - \tilde{W}_{i\zeta k}^T \Gamma_i^{-1} \tilde{W}_{i\zeta k} \\ &= -\left(\frac{1}{\tau} - \frac{a_i^2}{\tau} - a_i^2 \gamma_{ik} \right) e_{ik}^2 + \tau (1 + \tau \gamma_{ik}) \bar{\omega}_{ik}^2 \end{aligned}$$

$$\begin{aligned}
& -\tau(1-\tau\gamma_{ik})\left(\tilde{W}_{ik}^T S_{i\zeta k} - \frac{\gamma_{ik}}{1-\tau\gamma_{ik}}(a_i e_{ik} - \tau\bar{\omega}_{ik})\right)^2 \\
& + \frac{\tau\gamma_{ik}^2}{1-\tau\gamma_{ik}}(a_i^2 e_{ik}^2 + \tau^2 \bar{\omega}_{ik}^2) - \frac{2a_i e_{ik} \bar{\omega}_{ik}}{1-\tau\gamma_{ik}}. \quad (38)
\end{aligned}$$

Using the inequality $2a_i e_{ik} \bar{\omega}_{ik} \leq a_i^2 e_{ik}^2 + \bar{\omega}_{ik}^2$, it follows that:

$$\begin{aligned}
\Delta V_{ka} & \leq -\left(\frac{1}{\tau} - \frac{a_i^2}{\tau(1-\tau\gamma_{ik})}\right) e_{ik}^2 + \tau(1+\tau\gamma_{ik}) \bar{\omega}_{ik}^2 \\
& + \frac{\tau\gamma_{ik}^2}{1-\tau\gamma_{ik}}(\tau^2 \bar{\omega}_{ik}^2) + \frac{a_i^2 e_{ik}^2 + \bar{\omega}_{ik}^2}{1-\tau\gamma_{ik}} \\
& \leq -\frac{1}{\tau}(1-a_i^2 \mu_1) e_{ik}^2 + \frac{1+\tau}{1-\tau\gamma_{ik}} \bar{\omega}_{ik}^2. \quad (39)
\end{aligned}$$

Based on the expression of V_{kb} in function (37), the forward difference of V_{kb} is derived as follows:

$$\begin{aligned}
\Delta V_{kb} & = V_{(k+1)b} - V_{kb} \\
& = \tilde{W}_{i\zeta(k+1)}^T P(k+1) \tilde{W}_{i\zeta(k+1)} - \tilde{W}_{i\zeta k}^T P(k) \tilde{W}_{i\zeta k} \\
& = a_i e_{ik}^T S_{i\zeta k}^T \Gamma_i P(k+1) a_i \Gamma_i S_{i\zeta k} e_{ik} \\
& + S_{i\zeta k}^T \Gamma_i^T \tau \bar{\omega}_{ik} P(k+1) \Gamma_i \tau S_{i\zeta k} \bar{\omega}_{ik} \\
& - 2a_i e_{ik}^T S_{i\zeta k}^T \Gamma_i P(k+1) (I - \tau \Gamma_i S_{i\zeta k} S_{i\zeta k}^T) \tilde{W}_{i\zeta k} \\
& + 2S_{i\zeta k}^T \Gamma_i \tau \bar{\omega}_{ik} P(k+1) (I - \tau \Gamma_i S_{i\zeta k} S_{i\zeta k}^T) \tilde{W}_{i\zeta k} \\
& - 2a_i e_{ik}^T S_{i\zeta k}^T \Gamma_i^T P(k+1) \Gamma_i \tau S_{i\zeta k} \bar{\omega}_{ik} \\
& + \tilde{W}_{i\zeta k}^T (I - \tau \Gamma_i S_{i\zeta k} S_{i\zeta k}^T)^T P(k+1) \\
& \cdot (I - \tau \Gamma_i S_{i\zeta k} S_{i\zeta k}^T) \tilde{W}_{i\zeta k} - \tilde{W}_{i\zeta k}^T P(k) \tilde{W}_{i\zeta k}. \quad (40)
\end{aligned}$$

It can be proved that when the condition (30) holds, $\|(I - \tau \Gamma_i S_{i\zeta k} S_{i\zeta k}^T) \tilde{W}_{i\zeta k}\| \leq \|\tilde{W}_{i\zeta k}\|$. Subsequently, combined with (24), it follows that:

$$\begin{aligned}
\Delta V_{kb} & \leq \delta a_i^2 \Gamma_i \mu_3^2 \|e_{ik}\|^2 + 2\mu_3 a_i \delta \|\tilde{W}_{i\zeta k}\| \|e_{ik}\| \\
& - \kappa \tau \lambda_1 \|\tilde{W}_{i\zeta k}\|^2 + 2\tau \delta \mu_3 \|\bar{\omega}_{ik}\| \|\tilde{W}_{i\zeta k}\| \\
& + 2a_i \delta \Gamma_i \tau \mu_3^2 \|e_{ik}\| \|\bar{\omega}_{ik}\| \\
& + \delta \mu_3^2 \Gamma_i \tau^2 \bar{\omega}_{ik}^2 \quad (41)
\end{aligned}$$

where $\mu_3 = \sup_{k \geq k_0} \|S_{i\zeta k}\|$. Finally, based on (39) and (41), the difference of Lyspunov function V_k is

$$\begin{aligned}
\Delta V_k & = \Delta V_{ka} + \Delta V_{kb} \\
& \leq -\left(\frac{1}{\tau} - \frac{a_i^2(1+\tau)}{\tau(1-\tau\gamma_{ik})} - \delta a_i^2 \Gamma_i \mu_3^2\right) e_{ik}^2 - (\kappa \tau \lambda_1) \|\tilde{W}_{i\zeta k}\|^2 \\
& + 2\delta \mu_3 a_i \|\tilde{W}_{i\zeta k}\| \|e_{ik}\| + 2\tau \delta \mu_3 \|\bar{\omega}_{ik}\| \|\tilde{W}_{i\zeta k}\| \\
& + 2a_i \delta \Gamma_i \tau \mu_3^2 \|e_{ik}\| \|\bar{\omega}_{ik}\| \\
& + \left(\frac{1+\tau}{1-\tau\gamma_{ik}} + \delta \mu_3^2 \Gamma_i \tau^2\right) \bar{\omega}_{ik}^2. \quad (42)
\end{aligned}$$

Using the Schwartz inequality and the inequality $2xy \leq x^2 + y^2$, we have

$$\begin{aligned}
2\delta \mu_3 a_i \|\tilde{W}_{i\zeta k}\| \|e_{ik}\| & \leq \frac{4\delta^2 a_i^2 \mu_3^2}{\kappa \tau \lambda_1} e_{ik}^2 + \frac{\kappa \tau \lambda_1}{4} \tilde{W}_{i\zeta k}^2 \\
2\tau \delta \mu_3 \|\bar{\omega}_{ik}\| \|\tilde{W}_{i\zeta k}\| & \leq \frac{4\tau \delta^2 \mu_3^2}{\kappa \lambda_1} \bar{\omega}_{ik}^2 + \frac{\kappa \tau \lambda_1}{4} \tilde{W}_{i\zeta k}^2 \\
\delta \Gamma_i \mu_3^2 (2a_i \tau \|e_{ik}\| \|\bar{\omega}_{ik}\|) & \leq \delta \Gamma_i \mu_3^2 a_i^2 e_{ik}^2 + \delta \Gamma_i \mu_3^2 \tau^2 \bar{\omega}_{ik}^2. \quad (43)
\end{aligned}$$

Note that $\gamma_{ik} \leq \Gamma_i \mu_3^2$. Thus,

$$\frac{1+\tau}{1-\tau\gamma_{ik}} \leq \mu_1 \quad (44)$$

where μ_1 is given in (34). By substituting the above inequalities in (42), we obtain

$$\begin{aligned}
\Delta V_k & \leq -\lambda_4 e_{ik}^2 \\
& - \frac{1}{2}(\kappa \tau \lambda_1) \|\tilde{W}_{i\zeta k}\|^2 + (\mu_1 + \mu_2) \bar{\omega}_{ik}^2 \quad (45)
\end{aligned}$$

where $\lambda_4 = 1/\tau(1 - a_i^2(\mu_1 + \mu_2/\tau))$, μ_1 is given in (34) and μ_2 is given in (35).

If the condition (31) holds and

$$\|\tilde{W}_{i\zeta k}\|^2 > \frac{2}{\kappa \tau \lambda_1} (\mu_1 + \mu_2) \omega_i^{*2}. \quad (46)$$

Then, we will obtain

$$\Delta V_k < 0. \quad (47)$$

This implies that $\tilde{W}_{i\zeta k}$ will converge into the compact set described by (32). Using $|\tilde{W}_{i\zeta k}^T S_{i\zeta k}| \leq \|\tilde{W}_{i\zeta k}\| \|S_{i\zeta k}\|$, the upper boundedness of the convergence error (33) can be obtained. This ends the proof. ■

Theorem 2: Consider the learning error system (12) which satisfies Assumption 1. If the conditions (30) and (31) hold, then the convergence speed (denoted as λ_v) for the learning error system (12) is derived as follows:

$$\lambda_v = \ln \frac{1}{1 - \lambda_5/\lambda_6} \quad (48)$$

where $\lambda_5 := \min\{\lambda_4, \kappa \tau \lambda_1/2\}$, $\lambda_6 = \max\{\tau^{-1}, (1 + \delta)\Gamma_i^{-1}\}$, λ_4 , λ_1 , δ , and κ are parameters given in Theorem 1.

Proof: Consider the Lyapunov function given by (37). Based on (24), we have

$$\lambda_7 (e_{ik}^2 + \|\tilde{W}_{i\zeta k}\|^2) \leq V_k \leq \lambda_6 (e_{ik}^2 + \|\tilde{W}_{i\zeta k}\|^2) \quad (49)$$

where $\lambda_7 = \min\{\tau^{-1}, (1 + \lambda_2)\Gamma_i^{-1}\}$, $\lambda_6 = \max\{\tau^{-1}, (1 + \delta)\Gamma_i^{-1}\}$.

Based on the result (45) in the proof Theorem 1, we can get

$$\begin{aligned}
\Delta V_k & \leq -\lambda_4 e_{ik}^2 - \frac{1}{2}(\kappa \tau \lambda_1) \|\tilde{W}_{i\zeta k}\|^2 \\
& + (\mu_1 + \mu_2) \bar{\omega}_{ik}^2. \quad (50)
\end{aligned}$$

Let $\lambda_5 := \min\{\lambda_4, \kappa \tau \lambda_1/2\}$. Then, we have

$$\Delta V_k \leq -\lambda_5 (e_{ik}^2 + \|\tilde{W}_{i\zeta k}\|^2) + (\mu_1 + \mu_2) \bar{\omega}_{ik}^2. \quad (51)$$

Based on (49), it follows that:

$$\Delta V_k \leq -\frac{\lambda_5}{\lambda_6} V_k + (\mu_1 + \mu_2) \bar{\omega}_{ik}^2. \quad (52)$$

Since $\Delta V_k = V_{k+1} - V_k$, so that

$$\begin{aligned}
V_{k+1} & \leq \left(1 - \frac{\lambda_5}{\lambda_6}\right) V_k + (\mu_1 + \mu_2) \bar{\omega}_{ik}^2 \\
& = e^{-\lambda_v} V_k + (\mu_1 + \mu_2) \bar{\omega}_{ik}^2 \quad (53)
\end{aligned}$$

where λ_v is given in (48). Then, it follows that:

$$V_k \leq e^{-\lambda_v(k-k_0)} V_{k_0}$$

$$+ \sum_{\sigma=k_0}^{k-1} e^{-\lambda_v(k-\sigma-1)} (\mu_1 + \mu_2) \bar{\omega}_{i\sigma}^2 \quad \forall k \geq k_0 + 1. \quad (54)$$

Based on (49), we have

$$\begin{aligned} \left(e_{ik}^2 + \|\tilde{W}_{i\zeta k}\|^2 \right) &\leq \frac{\lambda_6}{\lambda_7} e^{-\lambda_v(k-k_0)} \left(e_{ik_0}^2 + \|\tilde{W}_{i\zeta k_0}\|^2 \right) \\ &\quad + \frac{1}{\lambda_7} \sum_{\sigma=k_0+1}^{k-1} e^{-\lambda_v(k-\sigma)} (\mu_1 + \mu_2) \bar{\omega}_{i\sigma}^2 \\ &\quad \forall k \geq k_0 + 1. \end{aligned} \quad (55)$$

Therefore, the convergence speed for the learning error systems can be derived. This ends the proof. ■

Due to the impact of noise, NN weights can not converge into small neighborhoods of their ideal values. To overcome this problem, we take the mean of the estimation of NN weights as the learning results.

After NN weight estimation error $\tilde{W}_{i\zeta k}$ converge into the compact set $\Omega_{\tilde{W}_{i\zeta k}}$ [as given by (32)], taking

$$\bar{W}_{i\zeta} = \text{mean}_{k \in [k_a, k_b]} \tilde{W}_{i\zeta k} \quad (56)$$

where k_a and k_b represent a time interval after the convergence of the NN weight estimation, a constant vector of neural weights $\bar{W}_{i\zeta}$ can be obtained.

Therefore, constant RBF NNs $\bar{W}_{i\zeta}^T S_{i\zeta k}$ can be utilized to approximate the nonlinear function $\xi_i(x_k, u_k)$ along the system trajectory

$$\xi_i(x_k, u_k) = \bar{W}_{i\zeta}^T S_{i\zeta k} + \epsilon'_i(x_k, u_k) \quad (57)$$

where $\epsilon'_i(x_k, u_k)$ is the approximation error of the constant NNs.

Remark 4: Theorem 1 provides a detailed quantitative analysis of the convergence accuracy of the proposed learning-based SDFI method. In the following, we show that parameters in the learning accuracy formulas (33) can be derived from measurable signals. First, for a trajectory generated from a dynamical system (2), an RBF network can be constructed by selecting neurons along the system trajectory. Thus, the number of neurons Q_ζ and the vector $S_{i\zeta k}$ can be calculated. Then, according to (6), the parameters α_1 , α_2 , and δ can be also calculated. Based on (25), (26), and (27), the parameters κ , μ_3 , and λ_1 can be calculated. Finally, based on (34) and (35) and the upper boundedness of disturbance (36), the upper boundedness of the convergence error can be calculated based on (33). Therefore, the learning accuracy formulas (33) provided by Theorem 1 can be used to evaluate the identification accuracy of the learning-based SDFI method in practical applications.

Remark 5: According to (79), another way to calculate λ_1 can be given as follows:

$$\lambda_1 = \lambda_{\min} \left(\sum_{\sigma=k}^{k+\delta-1} \Phi^T(\sigma, k) S_{i\zeta k} S_{i\zeta k}^T \Phi(\sigma, k) \right) \quad (58)$$

where $\lambda_{\min}(\cdot)$ denotes the minimum eigenvalue of a matrix and $\Phi(\sigma, k)$ is given in (22). Compared with (27), the value of λ_1 given by (58) will give a better estimation of the convergence accuracy of the FI scheme.

Remark 6: To guarantee convergence of the NN weights, it is necessary to satisfy the PE condition. Nevertheless, the PE condition is very restrictive. To overcome this difficulty, we explore a deterministic learning method to relax the PE condition. In deterministic learning [21], [22], the neurons in RBF NNs are divided into two classes, primary neurons [as defined by (5)] and secondary neurons. Primary neurons represent those close to the trajectory, whose outputs dominate the output of the network, and secondary neurons represent those far from the trajectory, whose outputs are very small and can be omitted.

Moreover, it is proven that for the periodic/quasi-periodic trajectories, the regressor vector consisting of the primary neurons will satisfy the PE condition (as shown by Lemma 1). That is, a PE condition of a reduced dimension regressor vector is less restrictive. With the partial PE condition (a reduced dimension PE condition), weights of primary neurons will converge into small neighborhoods of their ideal values, and the local accurate approximation of unknown system dynamics and fault dynamics can be achieved. If the system trajectories are not periodic/quasi-periodic, then the system trajectory in the fault transience interval can be used to train the RBF NN repeatedly. In this way, the partial PE condition of the neurons along the system trajectory in the transience interval can still be satisfied.

Remark 7: In [37], [38], and [39], system identification using concurrent learning methods has been developed. Through using a memory stack of historical data to satisfy a verifiable rank condition, concurrent learning can guarantee the exponential convergence of the parameters without the PE condition. Compared with concurrent learning, the main advantage of deterministic learning is that it does not employ historical data, so that the computational complexity can be simplified. The main disadvantage of deterministic learning is that it can only achieve identification with local accuracy.

In addition, in recent years, a number of interesting results have been obtained by utilizing deterministic learning [40], [41], [42]. In [40], a knowledge fusion technique based on deterministic learning has been developed to capture a broader identification region. In [41], knowledge acquired from deterministic learning has been applied to the rapid classification issue for dynamical patterns consisting of sampling sequences in a relatively large-scale dynamical dataset. In [42], a deterministic learning-based spatiotemporal dynamics identification scheme has been developed for modeling the dynamical process of human brain activity in human-robot cognitive interaction. Compared with the above works, we investigate the learning accuracy and learning rate of deterministic learning, which can be used to improve the performance of the above works.

Remark 8: The proposed learning identification method can also deal with random faults. Random faults include the faults that occur randomly and the faults with random nonlinearity. For randomly occurring faults, if the fault trajectories are recurrent, then the partial PE condition of the RBF NN can be satisfied, and the fault dynamics can be accurately identified. If the system trajectories due to random faults are not recurrent, then the system trajectory in the transience interval can be

used to train the RBF NN repeatedly. In this way, the accurate NN approximation can also be achieved. For the faults with random nonlinearity, identification of the fault nonlinearity, which is described by a random sequence, can be achieved by combining the learning schemes proposed in this article and stochastic learning schemes. In particular, the partial PE condition of RBF NN for a random sequence can be investigated by using similar technology given in [22]. Then, exponential convergence can be analyzed by using adaptive identification algorithms for stochastic systems. Identification faults with random nonlinearity will be investigated in future work.

Remark 9: The proposed learning identification method can not only deal with the dynamic systems described by (1) but also can deal with the diverse dynamic systems, such as multimachine power systems [23], and the dynamical process of human brain activity [42].

IV. SIMULATION STUDIES

To illustrate the effectiveness of the proposed FI scheme, numerical simulations on the robot manipulator and a compressor system are performed.

A. FI in the Robot Manipulator

In this section, we will carry out simulation experiments with the robot manipulator.

The system equations are shown below

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= f_2(x) + \phi(x) + d(t)\end{aligned}\quad (59)$$

where $x = [x_1, x_2]^T$ is the state vector, representing the angular position and angular velocity of the manipulator, $f_2(x) = -mgl \sin(x_1)/(2M) + u/M$ is the system dynamics, M is the inertia, g is the acceleration of gravity, and m and l are the mass and length of the link, respectively. $\phi(x) = \theta x_2$ is the fault item and $d(t)$ represents the zero-mean Gaussian-distributed noise. u is the control input which make the state x_1 track a reference signal $x_{1d}(t)$.

In the simulation, the parameters are specified as follows: $m = 0.5$, $g = 9.8$, $l = 0.5$, $M = 0.1$, and $\theta = 0.5$. The system uncertainty is assumed to be a 10% inaccuracy of the parameter m . The amplitude of Gaussian white noise is set to 0.05. The reference signal is chosen as $x_{1d}(t) = \cos(20\pi t)$. The sampling period is chosen as $\tau = 0.005$ s.

Now we study the identification of unknown dynamic $\xi(x_{1k}, x_{2k})$, which consists of the fault dynamics and system uncertainty. According to (7), the following learning estimator can be defined as:

$$\begin{aligned}\hat{x}_{2(k+1)} &= x_{2k} + a_i(\hat{x}_{2k} - x_{2k}) \\ &+ \tau(f_2(x_{1k}, x_{2k}) + \hat{W}_{2\zeta}^T S_{2\zeta}(x_{1k}, x_{2k}))\end{aligned}\quad (60)$$

where $a_i = 0.005$, x_{2k} denotes the sampled states and $\hat{x}_{2k}$ represents the state generated by learning estimator. $\hat{W}_{2\zeta}^T S_{2\zeta}(x_{1k}, x_{2k})$ is used to learn the unknown dynamics $\xi(x_{1k}, x_{2k})$. The RBF network employed in the simulation is constructed with the neurons set along the NN input trajectory

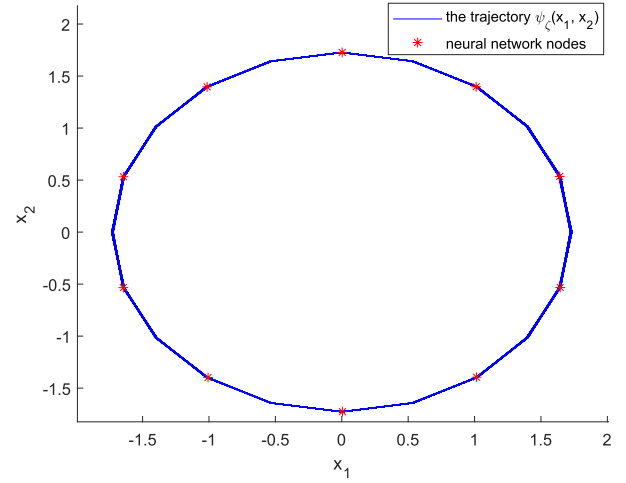


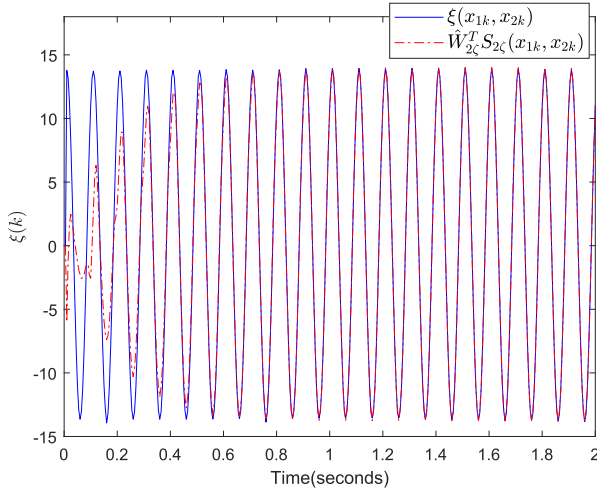
Fig. 1. NN nodes chosen along the NN input trajectory $\psi_\zeta(x_{1k}, x_{2k})$.

$\psi_\zeta(x_{1k}, x_{2k})$, which is illustrated in Fig. 1. The parameters of NN are chosen as: the number of neurons $Q_\zeta = 10$, the widths of neurons $\eta_i = 0.5$, the distance between two closest neuron centers $d_\zeta = 0.4$. The learning gain in (10) is chosen as $\Gamma_i = 30$. The results of learning the unknown dynamic $\xi(x_{1k}, x_{2k})$ are shown in Fig. 2. From Fig. 2(a), it can be seen that the NN outputs can approximate the system dynamic $\xi(x_{1k}, x_{2k})$.

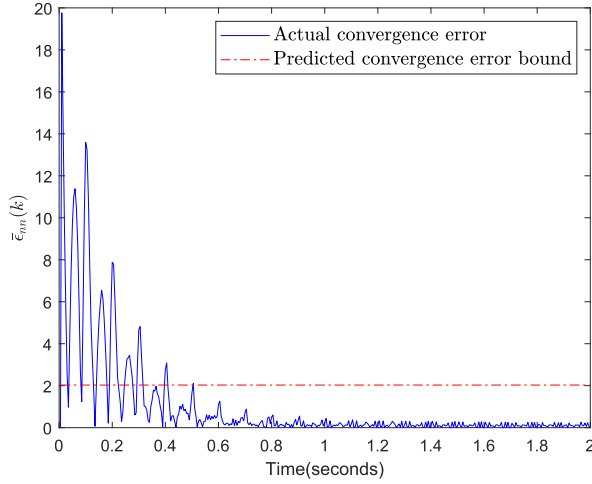
Based on the system trajectory and NN, the RBF vector $S_{2\zeta}(x_{1k}, x_{2k})$ can be calculated, and the parameter δ in (58) is chosen as 20. According to (58), λ_1 can be calculated as 0.3679. Moreover, based on (25)–(28), the parameters in (33) can be calculated as follows: $\omega_i^* = 0.05$, $\mu_3 = 1.1$, $\kappa = 1.85$, $\mu_1 = 1.18$, and $\mu_2 = 11.78$. Therefore, by employing (33), the convergence error bound is calculated as 2. The actual convergence error $\bar{\epsilon}_{nn}$ is obtained by (29). In Fig. 2(b), the convergence error bound and actual convergence error $\bar{\epsilon}_{nn}$ are shown. The simulation results show that the convergence error bound given by (33) can be calculated from measurable signals, and can provide an effective estimation of the actual convergence error.

In order to compare the proposed method with the recently developed FD techniques and evaluate its performance under noisy conditions, the following simulation experiments are conducted. In this experiment, the amplitude of Gaussian white noise is set to 4, and the finite-time concurrent learning (FTCL) and fixed-time concurrent learning (FxTCL) methods are employed to identify faults with the following learning laws, respectively,

$$\begin{aligned}\hat{\Theta}(k+1) &= \hat{\Theta}(k) - \Gamma_{FT} \left[\Xi_G \bar{d}(k) e_k^T + \Xi_C \left(\sum_{h=1}^P \bar{d}(\tau_h) e_{hk}^T \right. \right. \\ &\quad \left. \left. + \frac{\sum_{h=1}^P \bar{d}(\tau_h) e_{hk}^T}{\beta + \left\| \sum_{h=1}^P \bar{d}(\tau_h) e_{hk}^T \right\|} \right) \right]\end{aligned}\quad (61)$$



(a)



(b)

Fig. 2. Learning of the fault dynamic with the noise amplitude 0.05. (a) Function approximation by the RBF network. (b) Predicted convergence error bound and actual convergence error.

$$\begin{aligned} \hat{\Theta}(k+1) &= \hat{\Theta}(k) - \Gamma_{FXT} \left[\Xi_G \bar{d}(k) e_k^T + \Xi_C \left(\sum_{h=1}^P \bar{d}(\tau_h) (l e_{hk}^T \gamma_1 + l e_{hk}^T \gamma_2) \right) \right]. \end{aligned} \quad (62)$$

In the simulation, the parameters in learning laws (61) and (62) are chosen as $\Gamma_{FT} = 0.02I$, $\Gamma_{FXT} = 0.02I$, $\Xi_G = 0.5I$, $\Xi_C = 0.5I$, $P = 20$, $\gamma_1 = 2/3$, and $\gamma_2 = 2$. Parameter in learning laws (10) is chosen as $\Gamma = 30$. The amplitude of Gaussian white noise is set to 4. The NNs used in FTCL and FxTCL are constructed in the same way in the proposed deterministic learning FI (DLFI) method. To make a clearer comparison of these methods, the norms of approximation errors $E_f(k)$ are

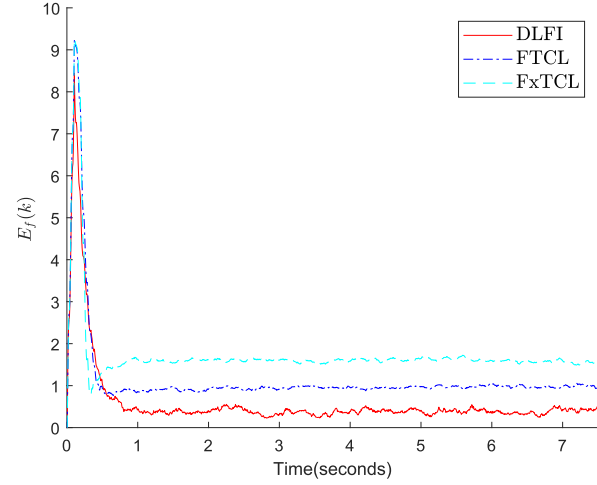


Fig. 3. Norms of approximation errors $E_f(k)$ using DLFI, FTCL, and FxTCL (with the noise amplitude 4).

employed, which are computed in the following way:

$$E_f(k) = \frac{1}{K} \sum_{\sigma=k-K+1}^k \bar{\epsilon}_{nn}(\sigma) \quad (63)$$

where $K = 20$. In Fig. 3, norms of approximation error $E_f(k)$ for DLFI, FTCL, and FxTCL are shown. It can be clearly observed that the convergence speed of DLFI is comparable to that of FTCL and FxTCL, while the convergence accuracy of DLFI is substantially higher than that of FTCL and FxTCL.

B. FI in the Mansoux Model

In this section, consider the Mansoux model, which describes the dynamic process of compressor rotating stall [43]

$$\begin{cases} E\dot{\phi} = -A\phi + \Psi_c(\phi) - T\bar{\psi} \\ \dot{\bar{\psi}} = \frac{1}{4l_c B^2} (S\phi - \Phi_T(\bar{\psi})) \end{cases} \quad (64)$$

where $\phi = [\phi_1, \phi_2, \dots, \phi_M]^T$ is the flow rate of M probes across the compressor with $M = 17$, $\bar{\psi}$ is the plenum pressure, E , A , S , and T are coefficient matrices, l_c is the total length of the compressor and ducts, B is a nondimensional parameter, $\Phi_T(\bar{\psi}) = (2\bar{\psi}/K_T)^{1/2}$ is the throttle characteristic function, and K_T is the throttle coefficient. $\psi_c(\phi)$ is the nonlinear compressor characteristic function obtained as follows:

$$\psi_c = \begin{cases} 12.117\phi^2 - 2.423\phi + 0.221 & \phi \leq 0.1 \\ -49.624\phi^3 + 39.509\phi^2 - 6.413\phi + 0.395 & 0.1 \leq \phi \leq 0.4 \\ -10.0695\phi^2 + 9.430\phi - 1.184 & \phi > 0.4. \end{cases} \quad (65)$$

The detailed parameters of the Mansoux model can be found in [43] and [44]. The Mansoux model can simulate the process of rotating stall by changing the throttle coefficient K_T . We choose $K_T = 9.37$ to generate one stall inception pattern, and the initial conditions are $\phi_i = 0.5595$, $\bar{\psi} = 0.9399$, the sampling period $\tau = 0.01$.

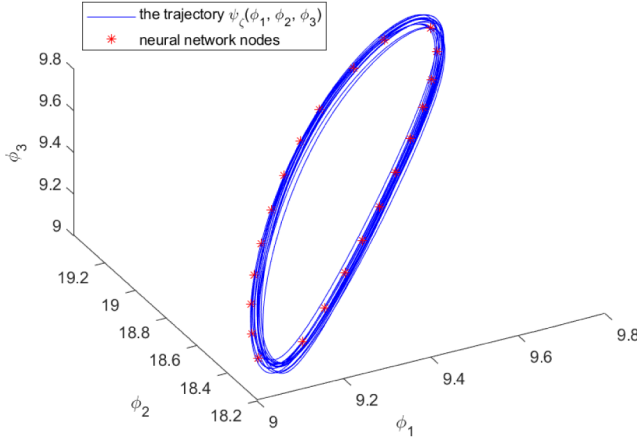


Fig. 4. NN nodes chosen along the NN input trajectory $\psi_\zeta(\phi_1, \phi_2, \phi_3)$.

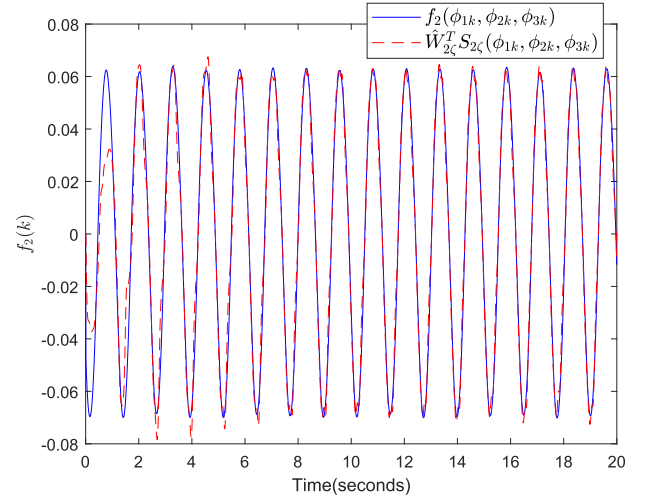
Now, we investigate how to achieve the identification of the system dynamics at the measurement point ϕ_i . In the following, the measurement point ϕ_2 is considered. Based on (64) and parameters in [43] and [44], the dynamic model for ϕ_2 can be described by

$$\dot{\phi}_2 = f_2(\phi_{2m}) + g_2(\phi_{2r}, \bar{\psi}) \quad (66)$$

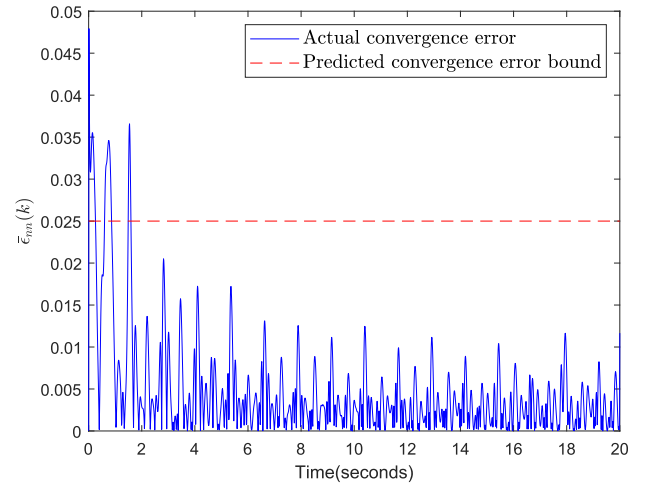
where $f_2(\phi_{2m}) = 0.8028\psi_c(\phi_2) - 0.1874(\psi_c(\phi_3) + \psi_c(\phi_1)) - 0.458(\phi_3 - \phi_1)$, $g_2(\phi_{2r}, \bar{\psi}) = 0.3042(\phi_4 - \phi_{17}) - 0.0577(\psi_c(\phi_4) + \psi_c(\phi_{17})) - 0.1835(\phi_5 - \phi_{16}) + 0.0368(\psi_c(\phi_5) + \psi_c(\phi_{16})) + 0.1553(\phi_6 - \phi_{15}) + 0.0186(\psi_c(\phi_6) + \psi_c(\phi_{15})) - 0.1249(\phi_7 - \phi_{14}) + 0.0144(\psi_c(\phi_7) + \psi_c(\phi_{14})) + 0.1149(\phi_8 - \phi_{13}) + 0.0091(\psi_c(\phi_8) + \psi_c(\phi_{13})) - 0.1050(\phi_9 - \phi_{12}) + 0.0080(\psi_c(\phi_9) + \psi_c(\phi_{12})) + 0.1023(\phi_{10} - \phi_{11}) + 0.0067(\psi_c(\phi_{10}) + \psi_c(\phi_{11})) - 0.1250\bar{\psi}$.

It is assumed the system dynamics $g_2(\phi_{2r}, \bar{\psi})$ is known while the system dynamics $f_2(\phi_{2m})$ is unknown. In the following, three states $\phi_{2m} = [\phi_1, \phi_2, \phi_3]^T$ are chosen to achieve the identification of the system dynamic $f_2(\phi_{2m})$. The RBF network employed in the simulation is constructed with the neurons set along the NN input trajectory $\psi_\zeta(\phi_1, \phi_2, \phi_3)$, which is illustrated in Fig. 4. The parameters of NN are chosen as: the number of neurons $Q_\zeta = 22$, the widths of neurons $\eta_i = 0.1$, and the distance between two closest neuron centers $d_\xi = 0.08$. The learning gain in (10) is chosen as $\Gamma_i = 0.035$. The initial weights of the RBF NN are $\hat{W}_{2\zeta 0}^T = 0$.

Based on the above system trajectory and NN, the RBF vector $S_{2\zeta}(\phi_1, \phi_2, \phi_3)$ can be calculated, and the parameter δ in (58) is chosen as 126. According to (58), λ_1 can be calculated as 0.84. Moreover, the parameters required in (33) can be calculated as follows: $\kappa = 1.88$, $\mu_1 = 1.15$, and $\mu_2 = 487.73$. The disturbance in the Mansoux model (66) is constructed by the zero-mean Gaussian distributed noise with the amplitude 0.00025. Thus, the upper boundedness of disturbance is chosen as $\omega_i^* = 0.00025$. Therefore, based on the above parameters and formula (33), the convergence error bound can be calculated as 0.025. In Fig. 5(b), the convergence error bound and actual convergence error are shown, which demonstrates that the convergence error bound given by (33)



(a)



(b)

Fig. 5. Identification of the rotating stall fault dynamic. (a) Function approximation by the RBF network. (b) Predicted convergence error bound and actual convergence error.

can provide an effective estimation of the actual convergence error.

According to (30), the range of the learning gain Γ_i is $(0, 82.64]$. In order to illustrate the effect of Γ_i on the learning accuracy and learning rate, the value of Γ_i changes in the range of $[0.001, 82.64]$, and the corresponding convergence error bound [as given by (33)] and convergence rate [as given by (48)] are shown in Fig. 6. The result shows that the ideal range for Γ_i that ensures both rapid convergence and high accuracy is $[1.5, 21]$.

In [26] and [27], the learning accuracy of deterministic learning has been analyzed. However, these analysis results are too complex and are difficult to evaluate the upper boundedness of the convergence bound of the NN approximation errors. In this article, the proposed learning accuracy formulas are simpler and are derived from measurable signals, so that

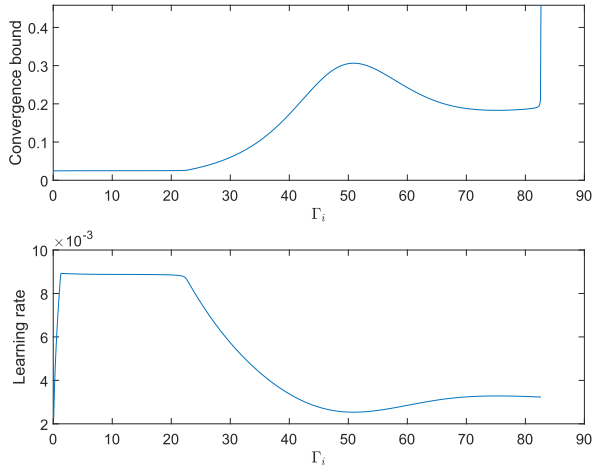


Fig. 6. Effect of Γ_i on the corresponding convergence error bound and corresponding rate.

the learning accuracy of the FI scheme can be effectively evaluated.

V. CONCLUSION

In this article, the problem of the identification of fault dynamics in SD nonlinear systems has been investigated, and the learning accuracy has also been analyzed. First, it is shown that the convergence of the learning error system can be turned into a stability problem for a class of SD LTV systems. Then, the exponential convergence property of the SD LTV system is derived by constructing a special TVPDM. Second, explicit formulas for learning accuracy are established, which reveal the relationship between system parameters and the convergence accuracy of the learning error system. The main advantage of the theoretical results given in this article is that the explicit formulas of the learning accuracy do not depend on any unknown parameters and thus can be accurately calculated.

The results presented in this article depend on the PE condition of RBF NNs. In future work, we will investigate the PE condition for other categories of NNs and extend the proposed method to these new network architectures.

APPENDIX A PROOF OF LEMMA 2

Proof: The proof of Lemma 2 is similar to the proof of [34, Lemma 2.5.2]. Consider the states x_k and w_k , corresponding to systems (13) and (14), respectively, with identical conditions $x_{k_0} = w_{k_0}$. Define $z_k = w_k - x_k$, $\Phi_{k,i} = A_{k-1}, \dots, A_i$, and $\Phi_{k,k} = I$. Then, we have

$$\begin{aligned} z_{k+1} &= A_k z_k + K_k C_k w_k \\ y'_k &= C_k w_k \end{aligned} \quad (67)$$

with the initial condition $z_{k_0} = 0$. The solution of the above system is

$$z_k = \sum_{i=k_0}^{k-1} \Phi_{k,i+1} K_i C_i w_i. \quad (68)$$

Let $e_i = K_i C_i w_i / |K_i C_i w_i|$ if $|K_i C_i w_i| \neq 0$ and $e_i = 0$ if $|K_i C_i w_i| = 0$. So that

$$\begin{aligned} |C_k (w_k - x_k)|^2 &= \left| \sum_{i=k_0}^{k-1} C_k \Phi_{k,i+1} K_i C_i w_i \right|^2 \\ &\leq \left(\sum_{i=k_0}^{k-1} |C_k \Phi_{k,i+1} e_i| |K_i| |C_i w_i| \right)^2 \\ &\leq \sum_{v=k_0}^{k-1} |C_v w_v|^2 \sum_{i=k_0}^{k-1} |C_k \Phi_{k,i+1} e_i|^2 |K_i|^2. \end{aligned} \quad (69)$$

Using the definition of the induced norm and the Schwartz inequality, we have

$$\begin{aligned} \left(\sum_{k=k_0}^{k_0+\delta} |C_k w_k|^2 \right)^{\frac{1}{2}} &\geq \left(\sum_{k=k_0}^{k_0+\delta} |C_k x_k|^2 \right)^{\frac{1}{2}} \\ &\quad - \left(\sum_{k=k_0+1}^{k_0+\delta} |C_k (w_k - x_k)|^2 \right)^{\frac{1}{2}}. \end{aligned} \quad (70)$$

So that, using (70), and the uniform observability of the original system

$$\begin{aligned} \left(\sum_{k=k_0}^{k_0+\delta} |C_k w_k|^2 \right)^{\frac{1}{2}} &\geq \sqrt{\beta_1} |w_{k_0}| \\ &\quad - \left(\sum_{k=k_0+1}^{k_0+\delta} \sum_{v=k_0}^{k-1} |C_v w_v|^2 \sum_{i=k_0}^{k-1} |C_k \Phi_{k,i+1} e_i|^2 |K_i|^2 \right)^{\frac{1}{2}} \\ &\geq \sqrt{\beta_1} |w_{k_0}| - \left(\sum_{v=k_0}^{k_0+\delta} |C_v w_v|^2 \right)^{\frac{1}{2}} \\ &\quad \left(\sum_{k=k_0+1}^{k_0+\delta} \sum_{i=k_0}^{k-1} |K_i|^2 |C_k \Phi_{k,i+1} e_i|^2 \right)^{\frac{1}{2}}. \end{aligned} \quad (71)$$

Changing the order of accumulation so that the accumulation in the last parenthesis becomes

$$\begin{aligned} &\sum_{k=k_0+1}^{k_0+\delta} \sum_{i=k_0}^{k-1} |K_i|^2 |C_k \Phi_{k,i+1} e_i|^2 \\ &= \sum_{i=k_0}^{k_0+\delta-1} \sum_{k=i}^{k_0+\delta} |K_i|^2 |C_k \Phi_{k,i+1} e_i|^2 \\ &\leq \sum_{i=k_0}^{k_0+\delta-1} \left(|K_i|^2 \sum_{k=k_0}^{k_0+\delta} |C_k \Phi_{k,i+1} e_i|^2 \right). \end{aligned} \quad (72)$$

Note that $k_0 + \delta - i \leq \delta$, $|e_i| = 1$, while $\Phi_{k,i+1} e_i$ is the solution of system (13) starting at e_i . Therefore, using the observability on the original system, and the condition on $K(\cdot)$, (72) becomes

$$\sum_{i=k_0}^{k_0+\delta-1} \left(|K_i|^2 \sum_{k=k_0}^{k_0+\delta} |C_k \Phi_{k,i+1} e_i|^2 \right) \leq k_\delta \beta_2. \quad (73)$$

Based on (71), we have

$$\left(\sum_{k=k_0}^{k_0+\delta} |C_k w_k|^2 \right)^{\frac{1}{2}} \geq \sqrt{\beta_1} |w_{k_0}| - \left(\sum_{v=k_0}^{k_0+\delta} |C_v w_v|^2 \right)^{\frac{1}{2}} \sqrt{k_\delta \beta_2}. \quad (74)$$

Then, it follows that:

$$\sum_{k=k_0}^{k_0+\delta} |C_k w_k|^2 \geq \frac{\beta_1}{(1 + \sqrt{k_\delta \beta_2})^2} |w_{k_0}|^2. \quad (75)$$

This ends the proof. ■

APPENDIX B PROOF OF LEMMA 4

Proof: 1) Consider the following system:

$$\begin{aligned} v_{k+1} &= v_k \\ y_{1k} &= S_{i\zeta k}^T v_k. \end{aligned} \quad (76)$$

Since $S_{i\zeta k}$ is PE, the system (76) is uniformly observable, that is,

$$\alpha_1 \|v_k\|^2 \leq \sum_{\sigma=k}^{k+\delta-1} y_{1\sigma}^2 = \sum_{\sigma=k}^{k+\delta-1} v_\sigma^T S_{i\zeta \sigma} S_{i\zeta \sigma}^T v_\sigma \leq \alpha_2 \|v_k\|^2. \quad (77)$$

Taking $K_k = -\tau \Gamma_i S_{i\zeta k}$, we have

$$\begin{aligned} & \sum_{\sigma=k}^{k+\delta-1} \|K(\sigma)\|^2 \\ &= \sum_{\sigma=k}^{k+\delta-1} \|\tau \Gamma_i S_{i\zeta \sigma}\|^2 \\ &= \tau^2 \Gamma_i^2 \text{tr} \left(\sum_{\sigma=k}^{k+\delta-1} S_{i\zeta \sigma} S_{i\zeta \sigma}^T \right) \\ &\leq Q_\zeta \tau^2 \Gamma_i^2 \alpha_2 \end{aligned} \quad (78)$$

where Q_ζ is the dimension of vector $S_{i\zeta k}$. Based on Lemma 2, we have with the output injection $K_k = -\tau \Gamma_i S_{i\zeta k}$, the system (19) is uniformly observable, i.e.,

$$\sum_{\sigma=k}^{k+\delta-1} y_\sigma^2 = \sum_{\sigma=k}^{k+\delta-1} w_\sigma^T S_{i\zeta \sigma} S_{i\zeta \sigma}^T w_\sigma \geq \lambda_1 \|w_k\|^2 \quad (79)$$

where $\lambda_1 = \alpha_1 / (1 + (Q_\zeta)^{1/2} \Gamma_i \tau \alpha_2)^2$.

Based on (21), we have

$$\begin{aligned} & w_k^T (G_k^T P(k+1) G_k - P(k)) w_k \\ &= w_k^T \left(G_k^T \sum_{\sigma=k+1}^{k+\delta} \Phi^T(\sigma, k+1) \Gamma_i^{-1} \Phi(\sigma, k+1) G_k \right. \\ & \quad \left. - \sum_{\sigma=k}^{k+\delta-1} \Phi^T(\sigma, k) \Gamma_i^{-1} \Phi(\sigma, k) \right) w_k. \end{aligned} \quad (80)$$

Note that $\Phi(\sigma, k)$ is the state transition matrix of (19). We have $\Phi(\sigma, k+1)G_k = \Phi(\sigma, k)$, $\Phi(\sigma, k)w_k = w(\sigma)$. So that

$$\begin{aligned} & w_k^T (G_k^T P(k+1) G_k - P(k)) w_k \\ &= w_k^T \left(\sum_{\sigma=k+1}^{k+\delta} \Phi^T(\sigma, k) \Gamma_i^{-1} \Phi(\sigma, k) \right. \end{aligned}$$

$$\begin{aligned} & \left. - \sum_{\sigma=k}^{k+\delta-1} \Phi^T(\sigma, k) \Gamma_i^{-1} \Phi(\sigma, k) \right) w_k \\ &= \sum_{\sigma=k+1}^{k+\delta} w_\sigma^T \Gamma_i^{-1} w_\sigma - \sum_{\sigma=k}^{k+\delta-1} w_\sigma^T \Gamma_i^{-1} w_\sigma \\ &= \sum_{\sigma=k}^{k+\delta-1} w_{\sigma+1}^T \Gamma_i^{-1} w_{\sigma+1} - \sum_{\sigma=k}^{k+\delta-1} w_\sigma^T \Gamma_i^{-1} w_\sigma \\ &= \sum_{\sigma=k}^{k+\delta-1} \left(w_\sigma^T (I - \tau \Gamma_i S_{i\zeta \sigma} S_{i\zeta \sigma}^T) \Gamma_i^{-1} \right. \\ & \quad \left. \cdot (I - \tau \Gamma_i S_{i\zeta \sigma} S_{i\zeta \sigma}^T) w_\sigma - w_\sigma^T \Gamma_i^{-1} w_\sigma \right) \\ &= \sum_{\sigma=k}^{k+\delta-1} - (2 - \tau S_{i\zeta \sigma}^T \Gamma_i S_{i\zeta \sigma}) \tau w_\sigma^T S_{i\zeta \sigma} S_{i\zeta \sigma}^T w_\sigma. \end{aligned} \quad (81)$$

Based on (79), it follows that:

$$w_k^T (G_k^T P(k+1) G_k - P(k)) w_k \leq -\kappa \tau \lambda_1 \|w_k\|^2 \quad (82)$$

where $\kappa = \min_{\sigma \geq k} (2 - \tau S_{i\zeta \sigma}^T \Gamma_i S_{i\zeta \sigma})$. Therefore, the inequality (23) can be obtained.

2) Consider the following system:

$$\begin{aligned} v_{k+1} &= v_k \\ y_{2k} &= v_k. \end{aligned} \quad (83)$$

It is obvious that the above system is uniformly observable, that is,

$$\delta \|v_k\|^2 \leq \sum_{\sigma=k}^{k+\delta-1} y_{2\sigma}^2 = \sum_{\sigma=k}^{k+\delta-1} v_\sigma^T v_\sigma \leq \delta \|v_k\|^2. \quad (84)$$

Taking $K'(k) = -\tau \Gamma_i S_{i\zeta k} S_{i\zeta k}^T$, we have

$$\begin{aligned} \sum_{\sigma=k}^{k+\delta-1} \|K'(\sigma)\|^2 &= \sum_{\sigma=k}^{k+\delta-1} \|\tau \Gamma_i S_{i\zeta \sigma} S_{i\zeta \sigma}^T\|^2 \\ &\leq \tau^2 \Gamma_i^2 \sum_{\sigma=k}^{k+\delta-1} \|S_{i\zeta \sigma} S_{i\zeta \sigma}^T\|^2 \\ &\leq \tau^2 \Gamma_i^2 \lambda_3^2 \delta \end{aligned} \quad (85)$$

where λ_3 is the maximum eigenvalues of $S_{i\zeta \tau} S_{i\zeta \tau}^T$ for all $\tau \geq k_0$. Thus, with the output injection $K'_k = -\tau \Gamma_i S_{i\zeta k} S_{i\zeta k}^T$, the system

$$\begin{aligned} w_{k+1} &= (I - \tau \Gamma_i S_{i\zeta k} S_{i\zeta k}^T) w_k \\ y_{3k} &= w_k \end{aligned} \quad (86)$$

is uniformly observable. Based on Lemma 2, we have

$$\begin{aligned} w_k^T P(k) w_k &= \sum_{\sigma=k}^{k+\delta-1} w(\sigma)^T \Gamma_i^{-1} w(\sigma) \\ &\geq \sum_{\sigma=k}^{k+\delta-1} \frac{1}{\Gamma_i} w^T(\sigma) w(\sigma) \\ &\geq \frac{\lambda_2}{\Gamma_i} \|w_k\|^2 \end{aligned} \quad (87)$$

where $\lambda_2 = \delta / (1 + \tau \Gamma_i \lambda_3 \delta)^2$.

According to (86), it can be proved that when the condition (30) holds, then $w_\tau \leq w_k \forall \tau > k$. Thus, we have

$$\begin{aligned} w_k^T P(k) w_k &= \sum_{\sigma=k}^{k+\delta-1} w(\sigma)^T \Gamma_i^{-1} w(\sigma) \\ &\leq \frac{1}{\Gamma_i} \sum_{\sigma=k}^{k+\delta-1} w(\sigma)^T w(\sigma) \\ &\leq \frac{1}{\Gamma_i} \sum_{\sigma=k}^{k+\delta-1} w_k^T w_k \\ &\leq \frac{\delta}{\Gamma_i} \|w_k\|^2. \end{aligned} \quad (88)$$

Based on (87) and (88), the inequality (24) can be obtained. This ends the proof. ■

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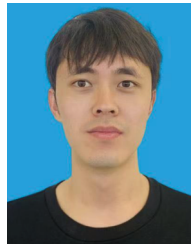
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